

UNIT II – BAYESIAN DECISION THEORY

Class 1: Bayesian Decision Theory, Bayesian Classification, Losses and Risks

Uncertainty



Probability



Decision



Best Action

 **Teacher Speaker Note:** Welcome students to Unit II. Explain that while Unit I focused on learning functions and representations from training data, Unit II addresses real-world machine intelligence: making the optimal choice when future observations are uncertain and mistakes have costs.

? Why Bayesian Decision Theory?

MOTIVATION

🔗 Real-World Scenario: Medical Diagnosis

A patient visits a clinic with common symptoms: **Fever, Cough, and Fatigue**.

Possible Outcomes: Flu, Bacterial Infection, or Common Cold.

⚠️ **Challenge:** The doctor is never 100% certain purely from symptoms.

Core Question: How can an AI or human make the best possible choice under uncertainty?

💡 The Bayesian Solution

Bayesian Decision Theory provides a principled mathematical framework when:

- Outcomes are **uncertain** (noisy real-world data).
- **Probabilities** can be quantified using prior knowledge and observed evidence.
- Different mistakes carry **different costs or penalties** (e.g., missing cancer vs. false alarm).

📄 Evidence / Data



% Probability



☰ Possible Outcomes



⚖️ Decision



✅ Optimal Action



Teacher Speaker Note: Ask students: "If you have a 90% chance of a mild cold and a 10% chance of an aggressive infection, do you just treat the cold because 90% is higher?" Emphasize that consequences matter just as much as probabilities!



Definition: Bayesian Decision Theory

EXAM ESSENTIAL

★ University Standard Definition

Bayesian Decision Theory is a fundamental statistical framework for making **optimal decisions under uncertainty** by combining probability information regarding state occurrences with the **losses or costs** incurred by taking particular actions.

1. Bayesian

Based on using prior knowledge and updating beliefs using newly observed evidence.

2. Uncertainty

The true state of nature cannot be determined with absolute 100% certainty.

3. Loss / Cost

The numerical penalty or disadvantage incurred if an incorrect decision is chosen.

Optimal Decision:

The specific choice among available actions that mathematically minimizes the expected loss (also termed minimizing risk).



In Simple Words: "Bayesian Decision Theory is our logical guide to choosing what action to take when we aren't completely sure what is true, but we know the odds and the penalties."



Teacher Speaker Note: Instruct students to write down this exact definition. Tell them that in BCA Semester exams, examiners check for 3 keywords: *Optimal decisions*, *Uncertainty*, and *Losses/Risk*.

Bayesian Decision-Making Idea

CORE CONCEPT

Observed Evidence



Class Probabilities



Cost / Loss Matrix



Calculate Expected Loss



Pick Lowest Risk

✗ Naive Decision: Most Probable Only

Ordinary classifiers merely ask: "Which class has the highest probability?"

Example:

- $P(\text{Disease A}) = 80\%$ (Mild Common Rash)
- $P(\text{Disease B}) = 20\%$ (Fatal Infection)

A naive model simply picks Disease A because $80\% > 20\%$.

✓ Bayesian Decision: Probability + Cost

Bayesian decision asks: "Considering the probabilities AND the cost of each possible mistake, what is safest?"

Cost Assessment:

- Missing Disease B causes patient death (Disaster).
- A Bayesian system can decide to test/treat Disease B despite its lower probability!

“ The highest probability event is NOT automatically the optimal decision when mistake consequences differ! ”



Teacher Speaker Note: Stress this comparison! A common student misconception is assuming that machine learning only means picking the maximum probability. Show them that in autonomous driving or medicine, high consequence always dictates decisions.

🔍 Essential Probability Basics

FOUNDATION

We only need three fundamental probability terms to master Bayesian Decision Theory:

1. Prior Probability

Our initial belief **before** observing any new sample data or symptom.

$$P(\text{Disease}) = 0.10 \text{ (10\%)}$$

2. Observed Evidence

The concrete feature measurements, test results, or symptoms collected from data.

$$x = \text{Fever, High BP}$$

3. Posterior Probability

The updated probability of a class **after** taking evidence into account.

$$P(\text{Disease} \mid \text{Symptoms}) = 0.70$$

</> Understanding the Vertical Bar Notation “ | ”

In mathematics, $P(A \mid B)$ translates in plain English to: *"Probability of event A occurring **given that** we already know event B happened."*

 **Teacher Speaker Note:** Emphasize that prior is "before the lab test" and posterior is "after the lab test". Explain that the vertical pipe | is not division—it means "conditional upon". Students frequently confuse this symbol in exams!



Understanding Bayes' Theorem

FORMULA

Bayes' Theorem provides the mathematical bridge to convert prior probabilities into updated posterior probabilities:

$$P(A|B) = \frac{P(B|A) \times P(A)}{P(B)}$$

P(A|B): Posterior — Prob. of class A given evidence B.

P(B|A): Likelihood — How likely evidence B is if class A is true.

P(A): Prior — Baseline probability of class A.

P(B): Evidence — Total marginal probability of observing B.



Golden Memory Formula for Exams: Posterior = (Likelihood × Prior) / Evidence



Teacher Speaker Note: Highlight that we do not need to derive Bayes' Theorem today. Tell students to memorize the word equation: Posterior = (Likelihood * Prior) / Evidence. This is a guaranteed 2-mark or 5-mark question in the semester exam.

√x Bayes' Theorem: Solved Example

NUMERICAL

🔖 Given Problem Data

Consider a disease screening scenario:

- $P(\text{Disease}) = 0.10$ (10% baseline population rate)
- $P(\text{Symptoms} | \text{Disease}) = 0.80$ (80% sensitivity)
- $P(\text{Symptoms}) = 0.20$ (20% of all people have symptoms)

Find: $P(\text{Disease} | \text{Symptoms})$

✍ Step-by-Step Solution

Step 1: Write Formula

$$P(\text{Disease} | \text{Symptoms}) = [P(\text{Symptoms} | \text{Disease}) \times P(\text{Disease})] / P(\text{Symptoms})$$

Step 2: Substitute Values

$$P(\text{Disease} | \text{Symptoms}) = (0.80 \times 0.10) / 0.20$$

Step 3: Compute Result

$$P(\text{Disease} | \text{Symptoms}) = 0.08 / 0.20 = \mathbf{0.40 (40\%)}$$

📌 Interpretation of Result:

Prior belief was only 10%. Observing the symptoms quadrupled our belief to 40%. The evidence substantially updated our confidence, demonstrating how Bayesian inference works!

👤 **Teacher Speaker Note:** Walk through the substitution explicitly on the board. Show students that $0.80 \times 0.10 = 0.08$, and $0.08 / 0.20 = 0.40$. Ask them: "Did the test give 100% certainty?" No! It updated probability from 10% to 40%.

What is Bayesian Classification?

CORE TOPIC

★ Textbook Definition

Bayesian Classification is a probabilistic machine learning approach that calculates the posterior probability of each possible class given an observed feature vector, assigning the input to the most appropriate class.

Input Features (x)



Calculate $P(C_i | x)$



Compare All Classes



Assign Winner Class

✉ Practical Example: Email Spam Filtering

Input Text: "Congratulations! You won cash prize."

Features: Words = {"Congratulations", "Won", "Cash", "Prize"}

- $P(\text{Spam} | \text{Email}) = 0.92$
- $P(\text{Not Spam} | \text{Email}) = 0.08$
- ✓ **Final Decision: Classify as SPAM**



Teacher Speaker Note: Tell students that Bayesian classification is the foundation of famous algorithms like the Naïve Bayes classifier. Emphasize that the model doesn't just guess "Yes/No", it outputs confidence scores for every category.



Bayesian Classification: Example

WALKTHROUGH



Clinical Patient Presentation

A patient reports to the diagnostic software with two measured binary features:

Observed Evidence (x):
[Fever = Yes, Cough = Severe]

The probabilistic ML model computes the posterior probability for all candidate classes.



Model Computed Posterior Table

Candidate Class (C_i)	Posterior: $P(C_i x)$	Status
Influenza (Flu)	0.75 (75%)	Highest
Common Cold	0.25 (25%)	Lower

Classification Rule: Choose $\text{argmax } P(C_i | x) \rightarrow \text{Predict FLU}$

⚠ Important ML Concept:
Probabilities are not arbitrarily invented by humans! They are calculated from historical training datasets using relative frequencies and probability density estimations.

 **Teacher Speaker Note:** Ask students: "If all we care about is minimizing classification error count, which class should we pick?" (Answer: Flu, 75%). Then use this as a bridge to slide 10: "Is Flu always the best medical action?"



Most Probable vs Best Decision

CRITICAL DISTINCTION

Dimension	Pure Classification	Bayesian Decision Theory
Core Question	"Which class is most likely to be true?"	"What action should I choose given probabilities & consequences?"
Assumed Loss	0–1 Loss (treats every error as equally bad)	Variable Loss (different mistakes have different costs)
Real Impact	Ignores severity of critical mistakes	Protects against catastrophic mistakes

 GOLDEN RULE: MOST PROBABLE ≠ ALWAYS THE BEST DECISION!

Suppose a fire alarm senses smoke. $P(\text{Toast Burning}) = 90\%$, $P(\text{House on Fire}) = 10\%$.
Even though Toast Burning is most probable, the *safest decision* is still to sound the fire alarm because the consequence of an ignored house fire is catastrophic.

 **Teacher Speaker Note:** This is a favorite university viva and theory question! Ask students: "Why can't self-driving cars just use simple classification?" Help them realize that a 2% chance of hitting a pedestrian carries infinitely higher cost than stopping unnecessarily.

Understanding Loss in Decisions

DEFINITION

★ University Definition

Loss, denoted as $\lambda(\alpha_i | C_k)$ or **L(action, state)**, is a formal numerical measure of the cost or penalty incurred by choosing action α_i when the true state of nature is actually class C_k .

Intuitive Meaning of Loss

- **Correct Action:** Zero loss or minimal cost (e.g., Loss = 0).
- **Minor Error:** Low loss (e.g., advising rest for healthy person, Loss = 1).
- **Critical Error:** Very high loss (e.g., sending sick patient home untreated, Loss = 100).

Illustrative Medical Loss Table

Actual State	Doctor's Action	Loss Value
Healthy	Decide Healthy	0 (No loss)
Healthy	Treat Disease	1 (Minor hassle)
Sick	Decide Healthy	100 (Disaster!)
Sick	Treat Disease	0 (Saved patient)

 **Key Takeaway:** Loss measures the real-life penalty of a single specific mistake. It quantifies "how bad was that decision?"

 **Teacher Speaker Note:** Point out row 3 of the table! Missing the disease has a loss of 100, while unnecessary medicine has a loss of 1. Because $100 \gg 1$, the penalty is heavily asymmetric.

🏠 The Loss Function & Matrix

MATHEMATICAL NOTATION

In textbook notation (Ethem Alpaydin, Ch. 3), loss is arranged in a structured **Loss Matrix**:

✓ Alpaydin Notation: λ_{ik}

$$\lambda_{ik} = \lambda(\alpha_i | C_k)$$

- α_i : Action taken (Decision i)
- C_k : True State of Nature (Class k)
- λ_{ik} : Penalty suffered when choosing α_i while true state is C_k .

🏠 Standard 2x2 Loss Matrix

Action Chosen (α)	Actual C_1 (Sick)	Actual C_2 (Healthy)
α_1 (Treat)	$\lambda_{11} = 0$	$\lambda_{12} = 2$
α_2 (Do Not Treat)	$\lambda_{21} = 50$	$\lambda_{22} = 0$

💡 Special Case: Zero-One (0-1) Loss

If all mistakes have identical cost (loss = 1 for any error, loss = 0 for correct classification), minimizing loss becomes identical to choosing the most probable class. This is called the **0-1 Loss Function**.

🗣️ **Teacher Speaker Note:** Show students how the indices work: λ_{12} means Action 1 chosen when Class 2 was true. Mention that Alpaydin uses Greek letter λ (lambda) for loss.



Understanding Risk (Expected Loss)

EXAM DEFINITION

★ Core Exam Definition

Risk, denoted as $R(\alpha_i | \mathbf{x})$, is the **expected loss** associated with taking a particular decision α_i , calculated by summing the losses over all possible true states weighted by their posterior probabilities.

$$R(\alpha_i | \mathbf{x}) = \sum_{k=1}^K \lambda(\alpha_i | C_k) \times P(C_k | \mathbf{x})$$

Class Probabilities $P(C_k | \mathbf{x})$

+

Loss Values $\lambda(\alpha_i | C_k)$

=

Expected Loss = RISK

Loss vs. Risk in One Line:

Loss is the cost of one known outcome.

Risk is the *average expected cost* taking uncertainty into account.

Bayes' Decision Rule:

Always select the action α^* that has the **minimum risk**:

$\alpha^* = \operatorname{argmin}_i R(\alpha_i | \mathbf{x})$



Teacher Speaker Note: Tell students: "Expected" in probability means weighted average! Risk is simply (Probability of event 1 * Loss of event 1) + (Probability of event 2 * Loss of event 2).



Step-by-Step Risk Calculation

NUMERICAL EXAMPLE

Given Probabilities: $P(\text{Disease } C_1 | x) = 0.30$ | $P(\text{Healthy } C_2 | x) = 0.70$

Decision 1: α_1 = "Treat Patient"

- Treat when Disease: Loss = 0
- Treat when Healthy: Loss = 2 (Medication cost)

Risk(Treat):

$$\begin{aligned} &= (0.30 \times 0) + (0.70 \times 2) \\ &= 0 + 1.4 = \mathbf{1.4} \end{aligned}$$

Decision 2: α_2 = "Do Not Treat"

- Don't Treat when Disease: Loss = 10 (Complication)
- Don't Treat when Healthy: Loss = 0

Risk(Do Not Treat):

$$\begin{aligned} &= (0.30 \times 10) + (0.70 \times 0) \\ &= 3.0 + 0 = \mathbf{3.0} \end{aligned}$$

🏆 Optimal Bayes Decision:

Since **Risk(Treat) = 1.4 < Risk(Do Not Treat) = 3.0**, the optimal decision is to **TREAT**.
Notice: Healthy is more likely (70%), but treating has lower expected risk!



Teacher Speaker Note: This slide proves everything we taught today! Remind students that simple classification would say "70% healthy, do nothing". But Bayesian decision theory calculates Risk = 1.4 vs 3.0 and chooses to treat.

Loss vs Risk & Class Summary

REVISION

Direct Exam Comparison

Loss (Cost)	Risk (Expected Loss)
Penalty for a single known outcome	Expected penalty over uncertain states
Does not use probabilities	Combines probabilities with losses
Notation: $\lambda(\alpha_i C_k)$	Notation: $R(\alpha_i x)$

Key Terms to Memorize for Exam

- **Bayesian Decision Theory:** Optimal choice under uncertainty.
- **Prior:** $P(C)$ before seeing evidence.
- **Posterior:** $P(C | x)$ after seeing evidence.
- **Bayes' Theorem:** $P(C|x) = [P(x|C) \times P(C)] / P(x)$.
- **Minimum Risk Rule:** Choose action with lowest $R(\alpha | x)$.

Evidence (x)



Posterior $P(C|x)$



Loss Matrix λ



Compute Risk $R(\alpha|x)$



Minimum Risk Action



Teacher Speaker Note: Have the students recite the pipeline once together before ending the class. Emphasize that in Class 2, we will explore Discriminant Functions and the famous Naïve Bayes Classifier!



Possible Examination Questions

QUESTION BANK

Short Answer (2 Marks)

- 1. Define Bayesian Decision Theory.
- 2. State Bayes' Theorem and write its formula.
- 3. Differentiate between prior and posterior probability.
- 4. Define 0–1 loss function.

Conceptual (5–7 Marks)

- 5. Explain the workflow of Bayesian classification with an example.
- 6. "Most probable class is not always the best decision." Explain with a scenario.
- 7. Clearly distinguish between Loss and Risk with suitable notations.

Numerical (5–8 Marks)

- 8. Given $P(A)=0.2$, $P(B|A)=0.8$, $P(B)=0.25$, calculate posterior $P(A|B)$.
- 9. Given a 2×2 loss table and posterior probabilities, calculate risk and state the optimal decision.



Self-Study Tip: Every single answer to these 9 questions is completely solved within Slides 1 to 15 of this presentation! Review the formulas and numerical tables before the exam.



Teacher Speaker Note: Advise students to practice writing question 7 and question 9 on paper as homework. This guarantees complete readiness for the internal assessment and university semester examinations.